higher education
& training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N5

8 APRIL 2019

This marking guideline consists of 8 pages.

QUESTION 1

A = $33,25_{10}$ = $100001,01_2$

B = $70,2_8$ = $111000,01_2$ {each octal is represented by 3 binary bits}

C = $B4,C_{16}$ = $10110100,11_2$ {each hexadecimal is represented by 4 binary bits}

1.1 A × C = $100001,01_2 \times 10110100,1100_2$

A × C = $100001,01_2 \times 10110100,1100_2$

$$\begin{array}{rl} 101101001100 & \\ \times \quad 10000101 & \\ \hline 101101001100 & ✓ \\ 10110100110000 & ✓ \\ 1011010011000000000 & ✓ \\ \hline 1011101111001,1111 & ✓ \end{array}$$

Therefore $101110111100,1111_2 = 13571,74_8$ ✓✓

(6)

1.2 B– $10,01_2$

B = $70,2_8$= $111000,01_2$

$$\begin{array}{lr} 111000,01_2 \xrightarrow{\hspace{6em}} & 111000,01_2 \\ 000010.01_2 \xrightarrow{1sc} \underset{✓}{111101,10} \xrightarrow{2sc} \underset{✓}{111101,10+1} \longrightarrow & \underline{111101,11_2} \quad ✓ \\ \text{Ignore the carry} \qquad & 1 \;\; 110110,11 \quad ✓ \\ & \underline{110110,00} \quad ✓ \end{array}$$

Therefore B-10,01 = $111000,01_2 - 10,01_2$ = 110110,00 = 54_{10} ✓

(6)

1.3 $C \div 11,1_2 = 101010100,1100_2 \div 11,1_2$
$= 10101010011,000_2 \div 1110_2$

```
                 0110011,110   ✓
            _________________
    1110   | 1011010011,000
              1110
             ________
              10001
               1110       ✓
             _______
             0011001
               1110
              ________
               10111      ✓
                1110
                _________
                 10010         ✓
                  1110
                  _________
                     10000     ✓
                     1110
                    _________
                      00100
```

Therefore $101010100,1100_2 \div 11,1_2 = 110011,101 = 63,A_{16}$ ✓

(6)
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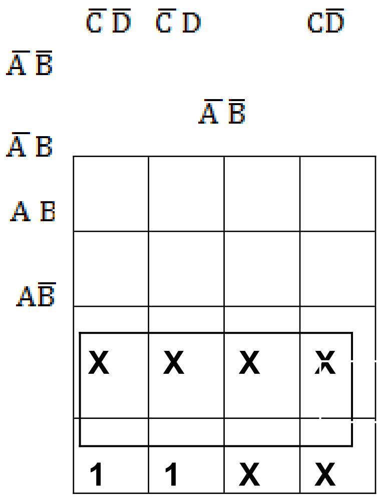
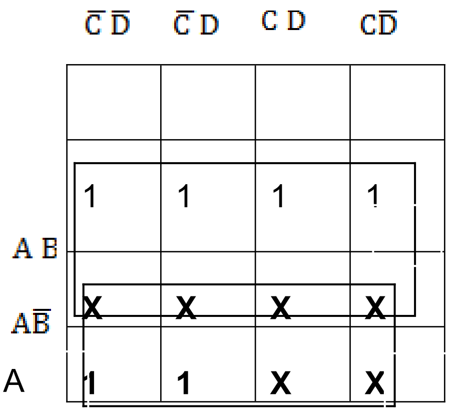
QUESTION 2

A	**B**	**C**	**D**	**F1**	**F2**	**F3**	**F4**	
0	0	0	0	0	0	0	0	✓
0	0	0	1	0	0	0	1	✓
0	0	1	0	0	0	1	1	✓
0	0	1	1	1	0	1	0	✓
0	1	0	0	0	1	1	0	
0	1	0	1	0	1	1	1	✓ ✓
0	1	1	0	0	1	0	1	✓
0	1	1	1	0	1	0	0	✓
1	0	0	0	1	1	0	0	✓
1	0	0	1	1	1	0	1	

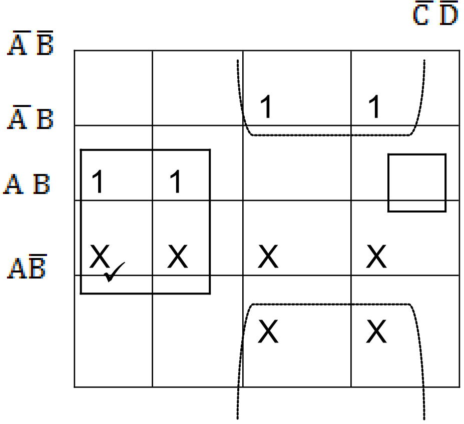
$$\left.\begin{array}{l}1010\\1011\\1100\\1101\\1110\\1111\end{array}\right\}\text{Don't cares}✓✓✓$$


$\overline{C}\,\overline{D}$
$\overline{C}\,D$
$C\overline{D}$
$\overline{A}\,\overline{B}$
$\overline{A}\,\overline{B}$
$\overline{A}\,B$
$A\,B$
$A\overline{B}$
X
X
X
X
1
1
X
X

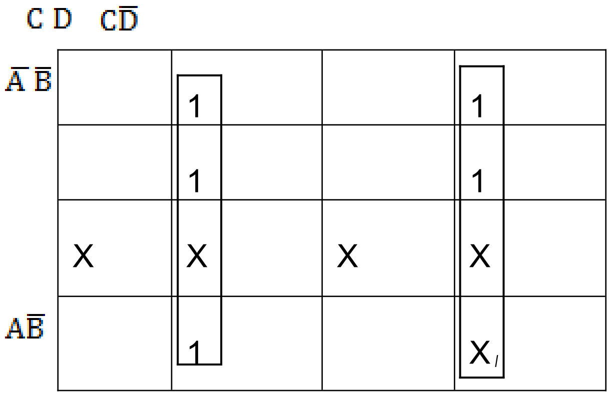
$F_1 = A$



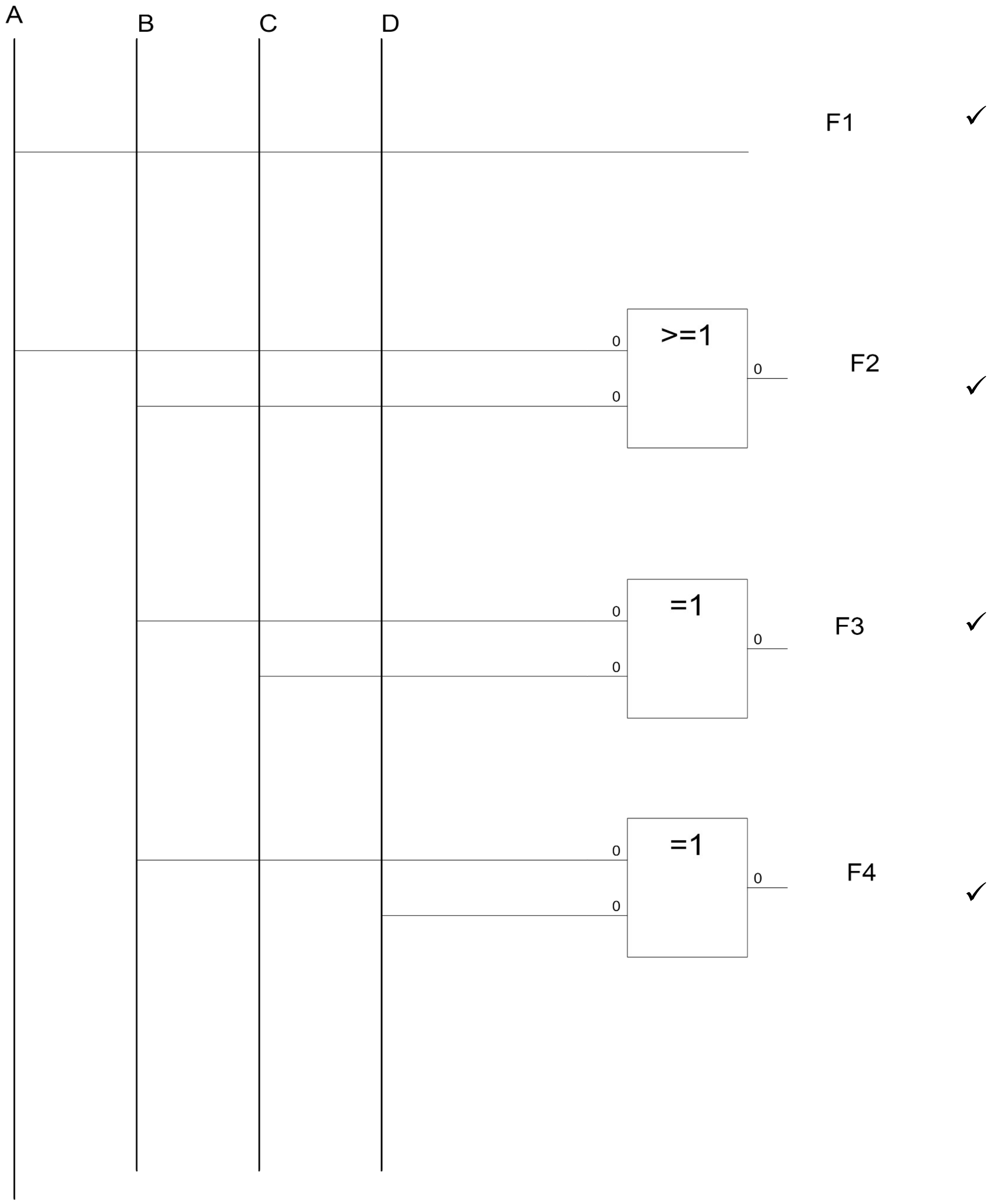
$F_2 = A + B$ √



$F_3 = B\overline{C} + \overline{B}C$✓



$F_4 = \overline{C}D + C\overline{D}$✓



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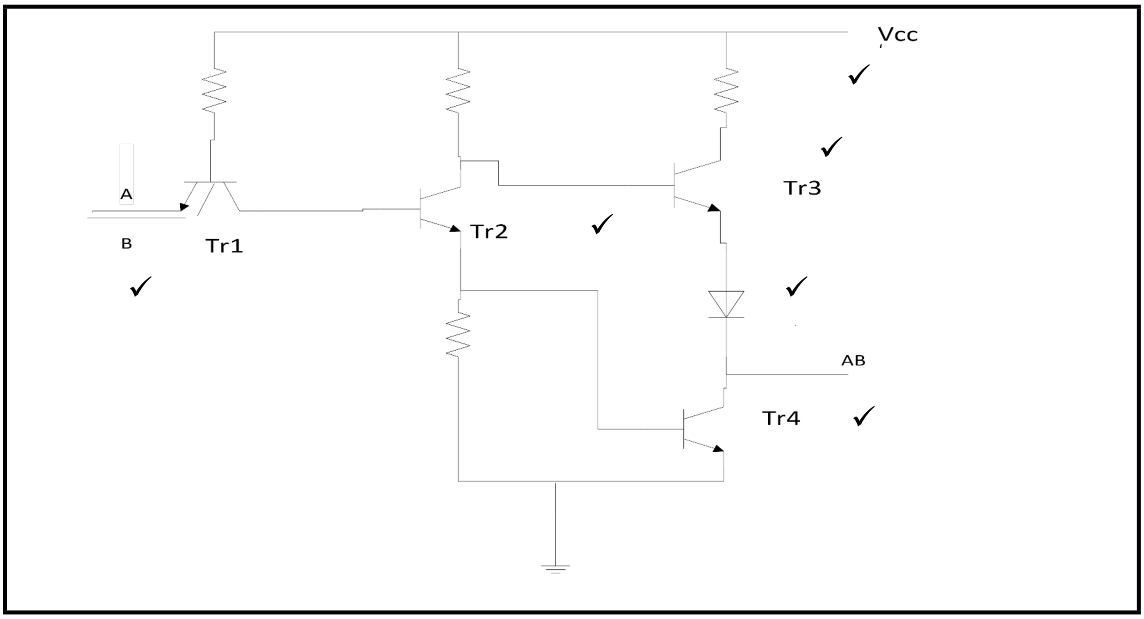
QUESTION 3

3.1



(6)

3.2 3.2.1 A
3.2.2 B
3.2.3 C
3.2.4 B

(4 × 1) (4)
[10]

QUESTION 4

4.1 4.1.1 Random Access Memory
4.1.2 Read Only Memory
4.1.3 Erasable Programmable Read Only Memory

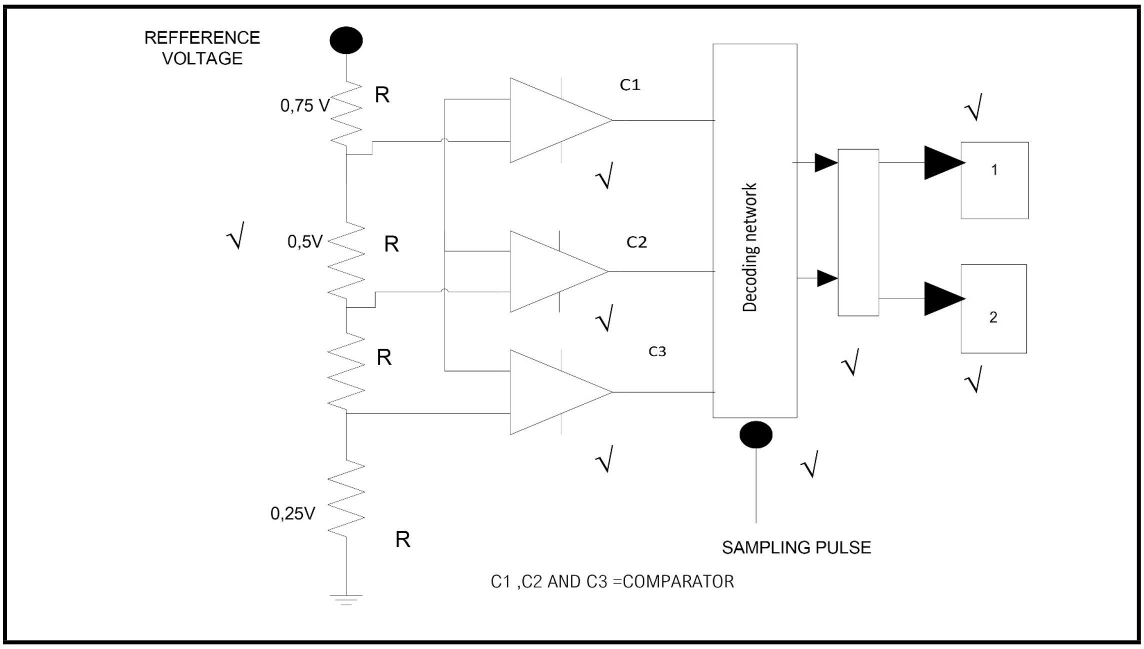
(3 × 2) (6)

4.2 4.2.1 True
4.2.2 False
4.2.3 False
4.2.4 False
4.2.5 False

(5 × 1) (5)
[11]

QUESTION 5

5.1



(10)

5.2 5.2.1 C
5.2.2 D
5.2.3 A
5.2.4 B

(4 × 1) (4)
[14]

QUESTION 6

6.1 6.1.1 Impact printers:

- Line printer
- Page printer
- Cylindrical printer
- Chain printer

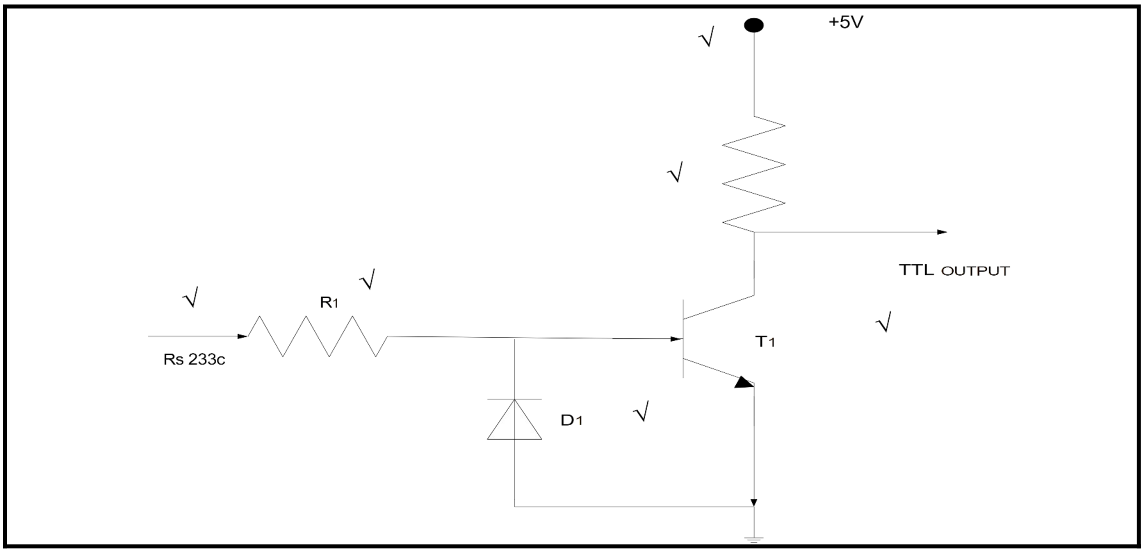
(Any 2 × 1) (2)

6.1.2 Non-impact printers:

- Electrostatic
- Electrothermal
- Ink jet
- Xerographic
- Laser
- Dot matrix
- Bubble

(Any 2 × 1) (2)

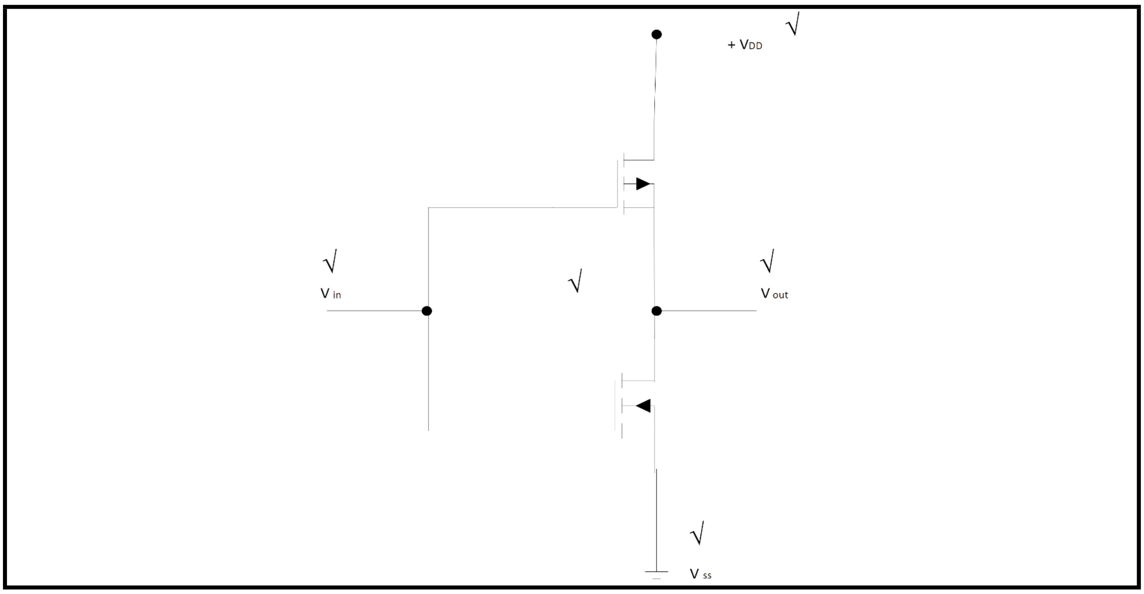
6.2



(6)
[10]

QUESTION 7

7.1



(5)

7.2
- Integrated circuits should be stored in aluminium foil or conductive foam (usually a dull black or pink).
- Ground to earth ground (the round prong of all properly wired outlets), the soldering-iron tip, the metal bench and the chassis of any test equipment used.
- In low-humidity, high-static conditions, connect your wrist to earth ground with a length of wire and a 1 MΩ series resistor.
- As with all integrated circuits, do not remove or insert MOS ICs with the power ON.

(4)
[9]

TOTAL: 100